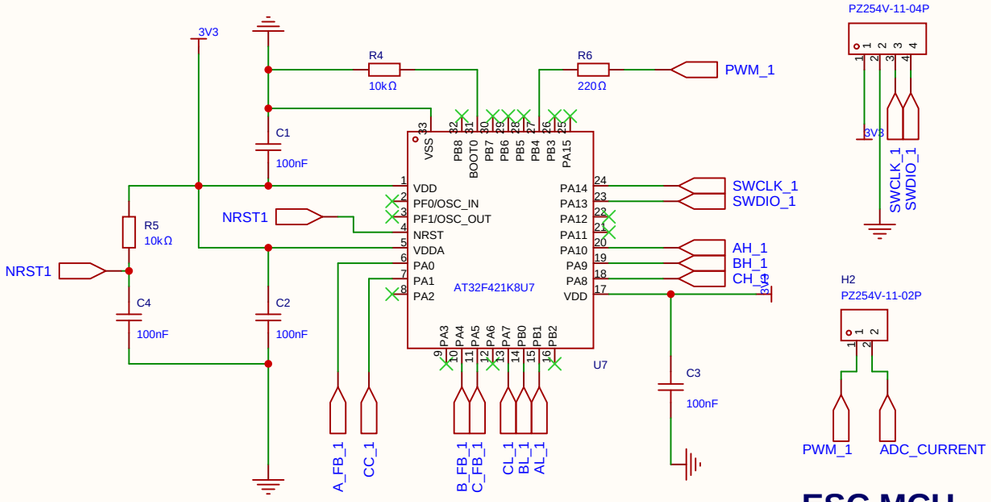
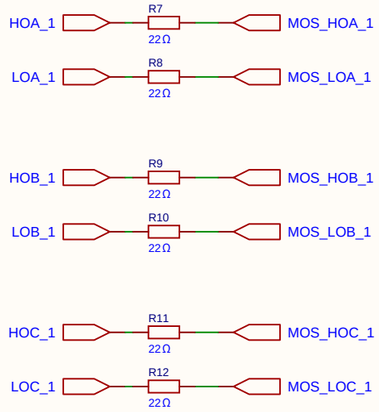
All decoupling capacitors and pull-up/down resistors may be substituted with smaller package sizes in subsequent revisions for space optimization

Component footprintss not specifically designated are assumed to be 0603 by default



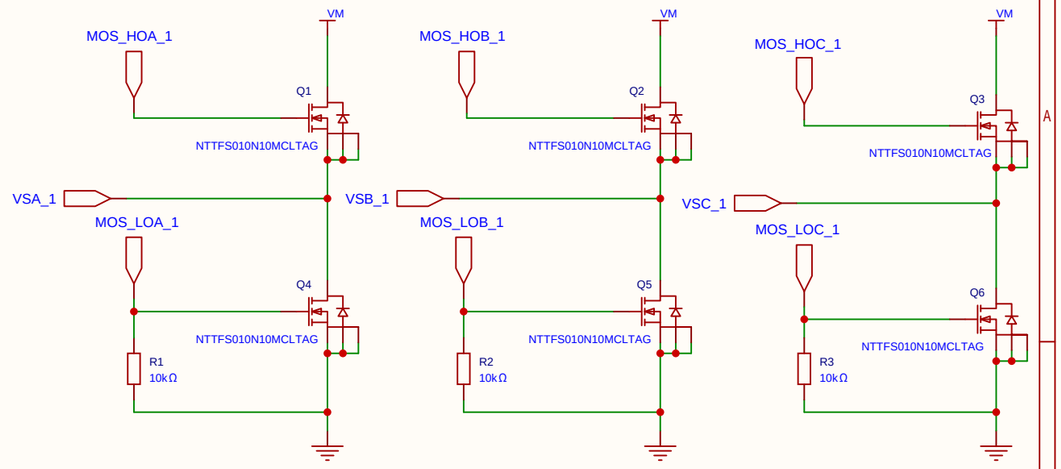
ESC MCU

Gate resistors can be substituted with smaller footprint later

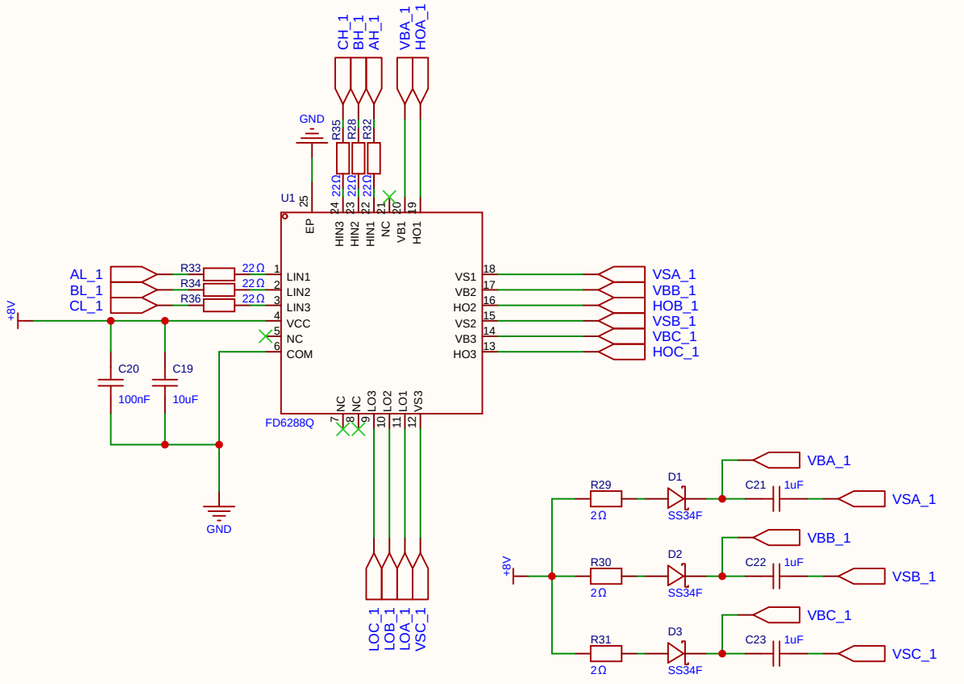


Gate Resistor

GaNFET G-S resistors can be substituted with smaller footprint later

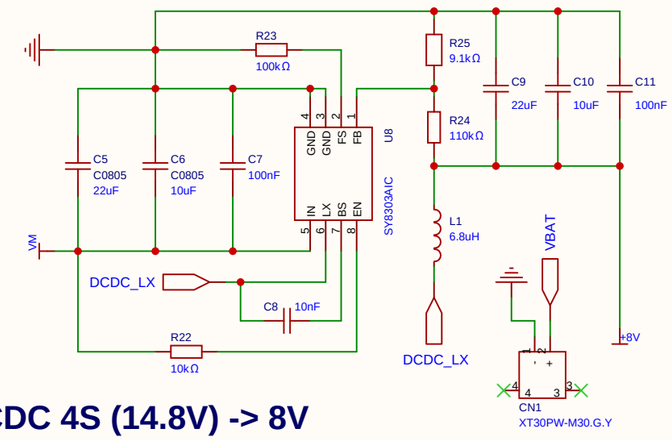


3-Phase Half-Bridge with Si MOSFETs



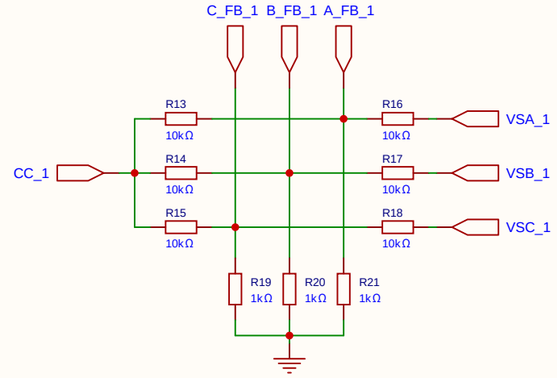
Si MOSFET 3-Phase Half-Bridge Driver

4S battery is stepped down to 8V by a DCDC converter to supply the LMG1210



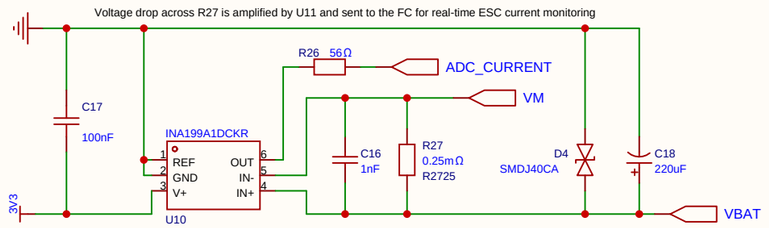
DCDC 4S (14.8V) -> 8V

0V pass detect resistors can be substituted with smaller footprint later



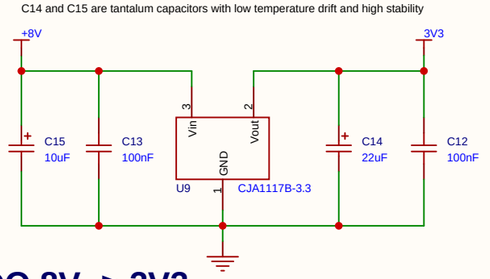
Back EMF Zero-Crossing Detection

Voltage drop across R27 is amplified by U11 and sent to the FC for real-time ESC current monitoring



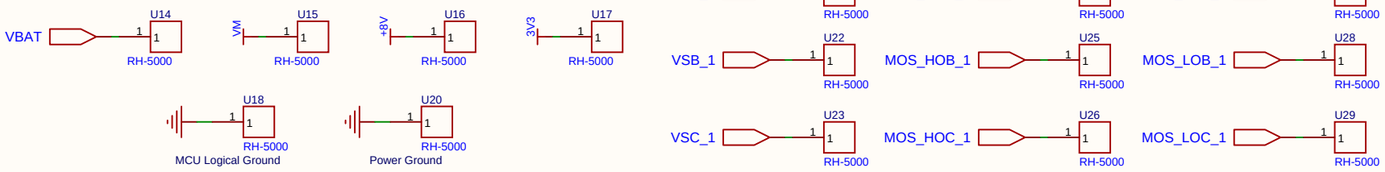
Current Sampling

C14 and C15 are tantalum capacitors with low temperature drift and high stability



LDO 8V -> 3V3

Test Point



原理图	Schematic1	创建日期	2026-03-28
板子	Phase 0: single MOSFET ESC verify	更新日期	2026-04-08
绘制	Zhiyi Li	图页	P1
审阅	Zhiyi Li	FYP: GaNFET based ESC-FC 2-in-1 PCB	
		版本 V1.0 / 尺寸 A4	页 1 共 1
嘉立创EDA		嘉立创EDA	